

7	(a)	(i)	1. The frequency of the source is the number of oscillations per unit time.	B1
			2. $n\lambda$	A1

		(ii)	<u>Either</u>	<u>Or</u>	
			$v = \frac{\text{distance}}{\text{time}} = \frac{n\lambda}{t}$	<i>f oscillation per unit time so $f\lambda$ is the distance per unit time</i>	M1
			Since $f = \frac{n}{t}$, $v = f\lambda$	<i>distance per unit time is v so $v = f\lambda$</i>	A1
	(b)	(i)	The incident wave from the loudspeaker and the reflected wave from the hard surface superpose / interfere / overlap. Since the two waves have the same speed and frequency, a stationary sound wave is formed		B1 B1 A0
		(ii)	The positions of the three nodes and two antinodes are where the intensity are zero and maximum respectively.		B1
		(iii)	Node to node = $\frac{1}{2} \lambda = 34 \text{ cm}$ $v = f\lambda$ $330 = f \times 0.68$ $f = 485 \approx 490 \text{ Hz}$		C1 A1
		(iv)	When $x = 20 \text{ cm}$, Intensity = $0.925 I$ When $x = 40 \text{ cm}$, Intensity = $0.25 I$ Since $I = kA^2$ $\frac{0.925I}{0.25I} = \frac{A_{x=20}^2}{A_{x=40}^2}$ $\frac{A_{x=20}}{A_{x=40}} = 1.92$		C1 C1 A1
	(c)	(i)	1. Each wavelength (of the white light) travels the same path difference or are in phase when reaching the zero order. All the wavelengths (of the white light) constructively interfere, producing a maximum. Hence the diffraction pattern has white light at the zero order		B1 B1 A0
			2. Since the wavelength of red light is longer than that of the blue light, the first order of red light is diffracted by a larger angle than that of the blue light since $\sin(\theta)$ is proportional to the wavelength. Thus, the first order of red light is further from the zero order than the first order of blue light.		B1 B1
		(ii)	$d \sin \theta = n \lambda$ $\frac{1}{N} \sin 61.0^\circ = 2 \times 625 \times 10^{-9}$ $N = 7.00 \times 10^5 \text{ line per m}$		C1 A1

